



Calculating theoretical probabilities from probability tree diagrams (one event)

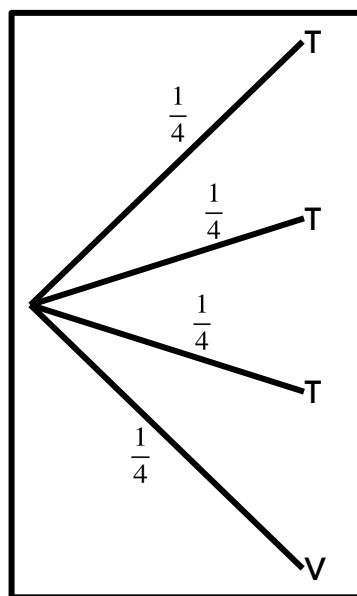
Task A: Probability trees to show outcomes

1) Both these **probability trees** show the outcomes and probabilities for the same trial (spinning a spinner).

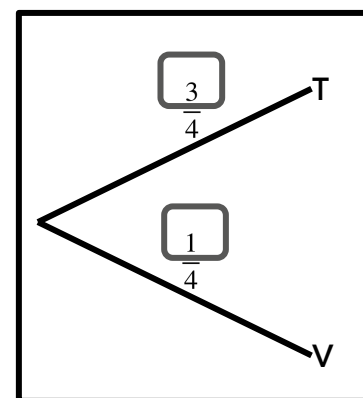
a) Use **probability tree 1** to complete **probability tree 2**.

b) Use these two **probability trees** to complete the following sentence:

“The probability of the spinner landing on a T is $\frac{3}{4}$ or 0.75 or 75%”.



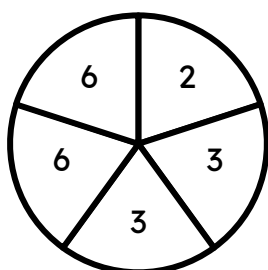
probability tree 1



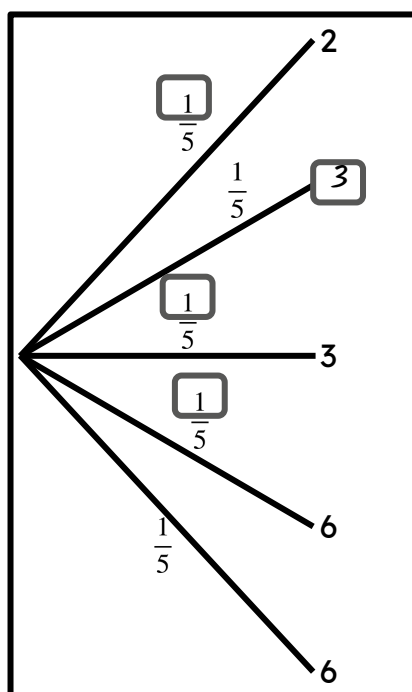
probability tree 2

2) Sofia will spin spinner A once.

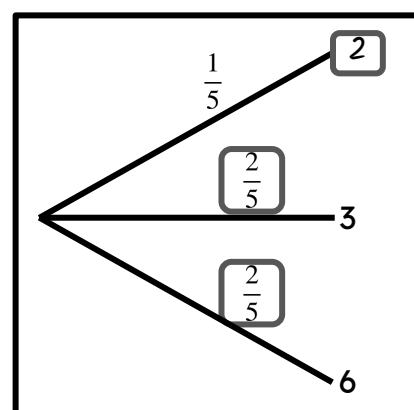
a) Complete the **probability tree** with five branches for this trial.



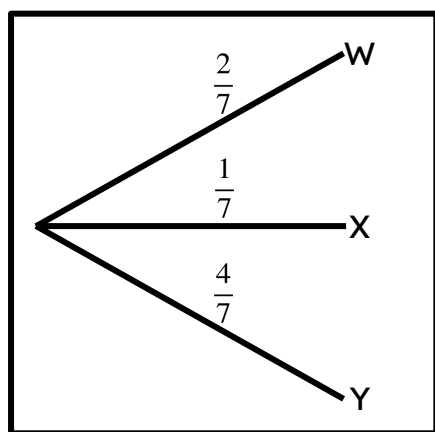
spinner A



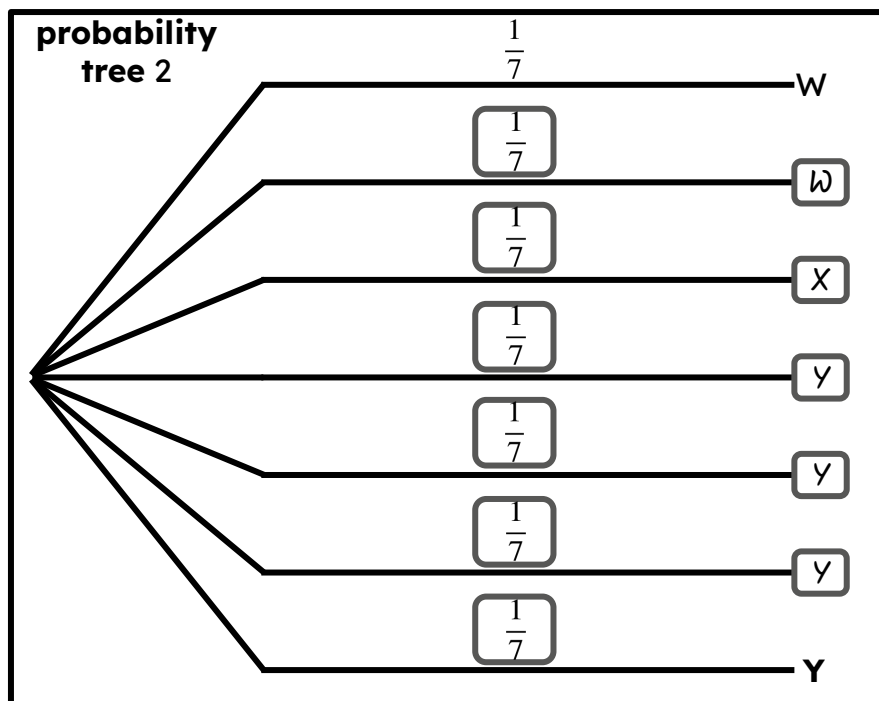
b) Hence, complete the **probability tree** which shows each of the unique outcomes for this trial.



3) Use **probability tree 1** to complete **probability tree 2**.

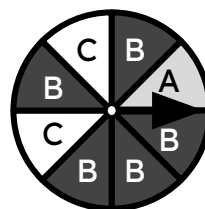


probability tree 1

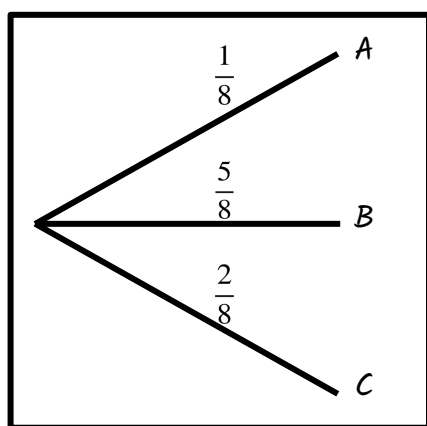


4) Jun will spin this spinner once.

a) By constructing an appropriate **probability tree**, list the probabilities of each unique outcome.

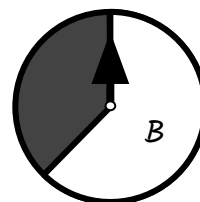
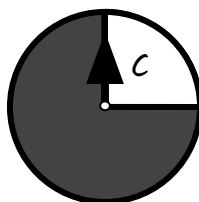
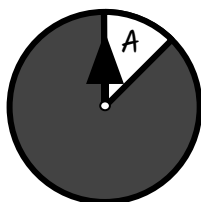


spinner



Whilst it is possible to draw an eight-branch **probability tree** to represent all eight sectors of the spinner, this isn't necessary (or practical) when listing the probabilities of the three unique outcomes: {A, B, C}.

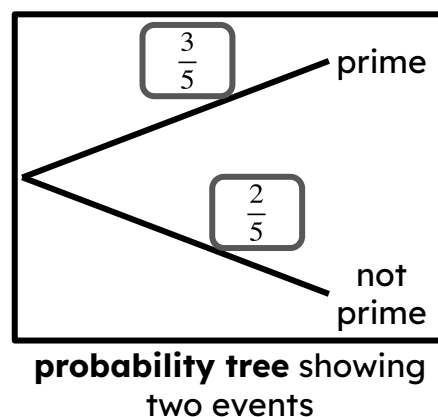
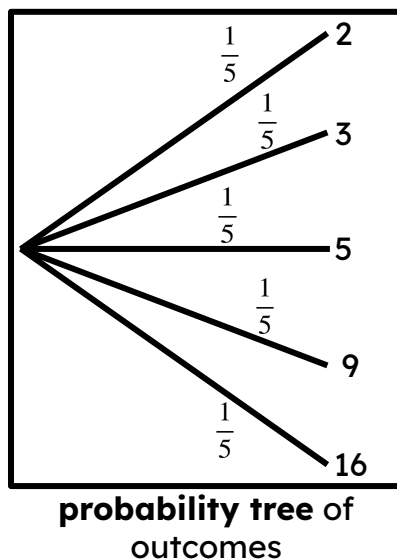
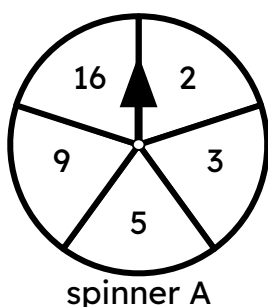
b) Mark the lighter segment on each of these outcome spinners with either A, B, or C to represent the probability that the spinner lands on that outcome.



Task B: Comparing trees that show events and outcomes

1) Izzy will spin spinner A once and look for the event that the spinner lands on a prime number. Izzy constructs a **probability tree** of outcomes.

Complete the **probability tree** showing two events: landing on a prime, or landing on a non-prime.



2) Aisha creates a **probability tree** of outcomes for a trial of spinning a spinner with 12 sectors.

a) Which of these outcomes are square numbers?

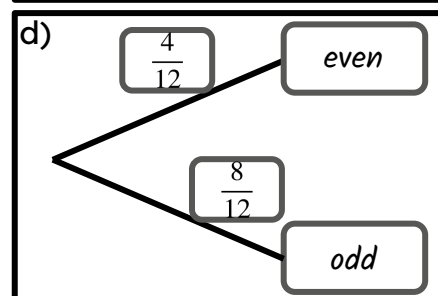
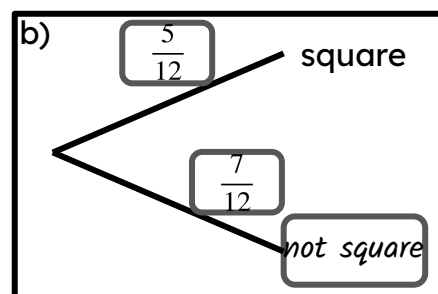
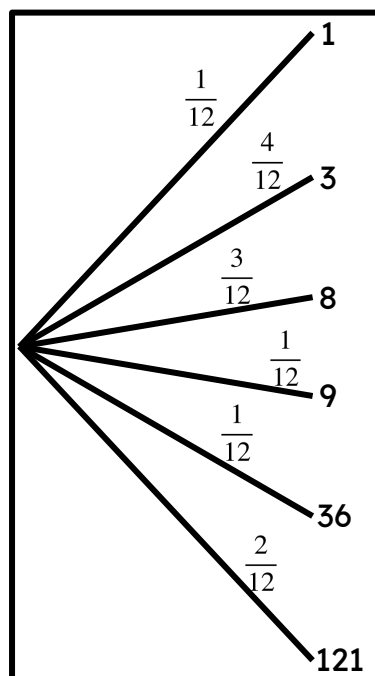
1, 9, 36, 121

b) Complete the **probability tree** of two events: landing on a square number, and landing on a non-square number.

c) Which of these outcomes are even numbers?

8, 36

d) Complete the **probability tree** of two events, where one event is the spinner landing on an even number.



Odd could also be not even.

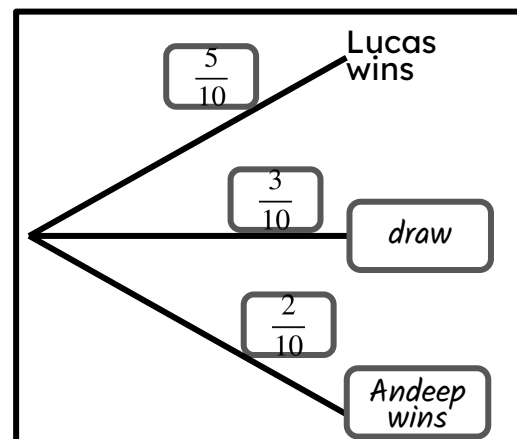
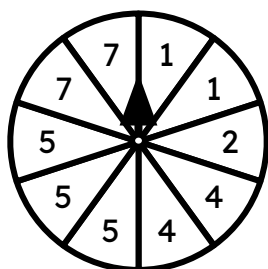


3) Lucas and Andeep play a game. Whoever gets the highest number wins, and if they get the same number, they draw.

Lucas spins the spinner, which lands on a 5.

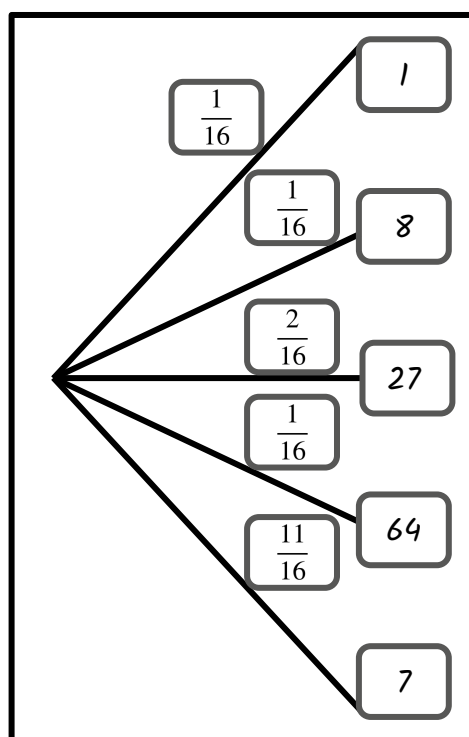
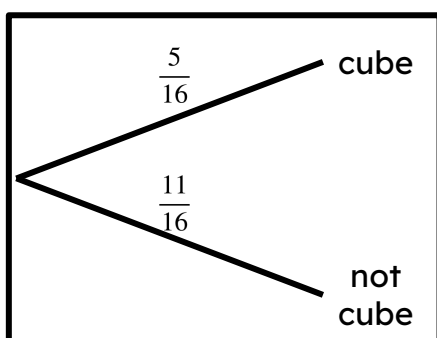
Andeep is about to spin the spinner.

Complete this **probability tree** to show the events of who could win.



4) Both **probability trees** represent the same trial.

Use the **probability tree** showing the probability of the event “landing on a cube” to create a possible **probability tree** of outcomes.



This is one example of what your probability tree may look like.

There must be at least one branch representing a cube number, and at least one branch representing a non-cube number.

Some examples of cube numbers include: 0, 1, 8, 27, 64, 125, 1000

